

Review

Flexible dry electrodes for electrocardiographic monitoring: Working principles, design strategies, and recent advances

Tianhang Liu,¹ Xili Hu,^{1,*} Dong Wang,³ Wenlu Shi,¹ Dongming Xing,¹ Chaoyu You,¹ Tailin Xu,² Mingwei Tian,¹ Lijun Qu,¹ and Xueji Zhang²

¹Cancer Institute, The Affiliated Hospital of Qingdao University, Research Center for Intelligent and Wearable Technology, College of Textiles and Clothing, Qingdao University, Qingdao, China

²School of Biomedical Engineering, Shenzhen University Health Science Center, Shenzhen, China

³Canada Qingdao Customs of the People's Republic of China, Qingdao, China

*Correspondence: huxili@qdu.edu.cn

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SUMMARY

The electrocardiogram (ECG) signal relies heavily on electrode performance, where flexible electronic materials play a crucial role. Flexible dry electrodes (FDEs), characterized by their skin-like Young's modulus and adhesion to biological interfaces, have emerged as an ideal solution for long-term, dynamic ECG monitoring. To fully explore their potentialities, an in-depth understanding of their working principles of both single-lead and multi-lead ECG monitoring systems is deeply examined, together with various ECG leads and electrode suitability. The design strategies for FDEs are analyzed, focusing on material selection and structural optimization and their impact on electrode performance. Furthermore, cutting-edge advances, such as e-skin/tattoo electrodes, textile-based electrodes, and microarray electrodes, are highlighted, emphasizing their clinical and practical significance. Finally, challenges in signal stability, environmental adaptability, and long-term usability are discussed, and the future directions for next-gen ECG monitoring solutions are proposed. This review aims to outline a roadmap for developing high-performance, user-centered ECG electrodes.

INTRODUCTION

As the centrepiece of the human circulatory system, the heart plays a crucial role. While blood is pumped to deliver oxygen and nutrients throughout the body, it also generates Electrocardiogram (ECG) signals through its beats, which are key indicators of cardiac health. During heart activity, the flow of ions in and out of cardiac muscle cells, particularly the transmembrane movement of sodium (Na⁺) and potassium (K⁺), creates a potential difference, which can be detected as a weak change in electric current on the surface of the skin.¹ Rather than having a specific magnitude, these current changes are also directional, such as vectors in physics, and are therefore called ECG vectors. The acquisition of an ECG relies on the accurate placement of electrodes on the surface of the skin, which are connected by wires to the positive and negative terminals of an ECG machine to form a closed circuit that collects the ECG signals. ECG leads—the electrical circuits formed between the electrodes—are able to detect electrical activity from different parts of the body, depending on the configuration and position of the electrodes, and synthesize these signals to obtain 3-dimensional (3D) ECG vectors.^{2,3} These vectors are projected in different planes, culminating in an ECG, which provides a visual image of the heart's function in the form of P-Q-R-S-T waveforms, and the detailed interpretation of these waveforms is essential for identifying func-

tional cardiac pathologies such as myocardial injury, arrhythmia, cardiac hypertrophy, and other cardiac pathologies. This provides clinicians with a potent tool for in-depth understanding and analysis of cardiac conditions.^{4,5} With advances in technology, ECG monitoring has evolved from traditional static monitoring to modern dynamic and wearable monitoring, and these innovations have dramatically improved the ease and accuracy of diagnosing cardiac disease and monitoring patients.

Under the conventional method, the acquisition of ECG signals involves the adhesion of a layer of conductive hydrogel to the skin in conjunction with the use of silver (Ag)/silver chloride (AgCl) noninvasive electrodes. This hydrogel would serve not only to immobilize the electrodes but also to reduce the contact impedance between the electrodes and the skin, thus ensuring high fidelity and stability of the collected ECG signals, which is ideally suited for static ECG monitoring; in clinical practice, this configuration is often used together with a 12-lead system for exhaustive cardiac function assessment.^{1,4–6} However, in regard to long-term monitoring, the gradual dehydration of the hydrogel layer can lead to poor electrode–skin contact, increasing impedance, and potentially significant degradation in signal quality.^{7–10} In addition, for people with more sensitive skin, prolonged hydrogel contact may cause discomfort, such as itching and redness, and, in some cases, even lead to allergic reactions. When ambulatory ECG monitoring is needed, the numerous electrodes and



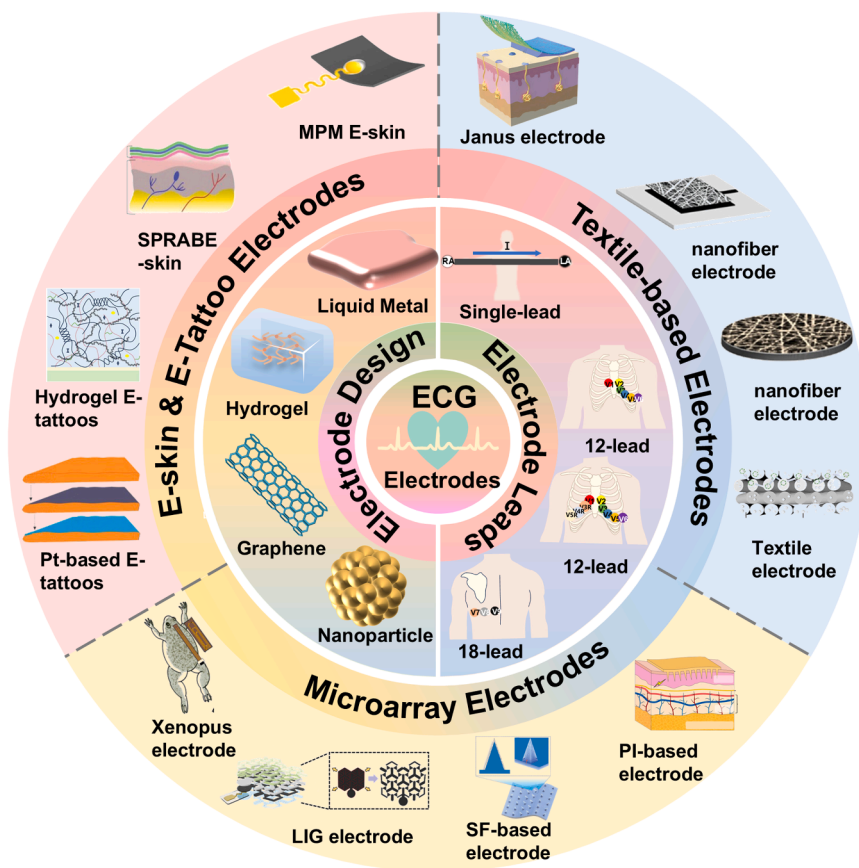


Figure 1. FDEs for ECG monitoring

From inside to outside, with schematic diagrams of ECG monitoring electrodes, electrode design strategies, and electrode leads, as well as advanced electrodes for ECG monitoring, including e-skin and e-tattoos (from left to right, Reproduced with permission from¹⁷; Copyright 2021 American Chemical Society. Reproduced with permission from¹⁸; Copyright 2024, Author(s). Reproduced with permission from¹⁹; Copyright 2023 Wiley-VCH. Reproduced with permission from²⁰; Copyright 2022 Elsevier), textile-based electrodes (bottom to top, Reproduced with permission from²¹; Copyright 2022 Wiley-VCH. Reproduced with permission from²²; Copyright 2021 American Chemical Society. Reproduced with permission from²³; Copyright 2022 American Chemical Society. Reproduced with permission from²⁴; copyright 2023 Elsevier Ltd.) and microarray electrodes (from left to right, Reproduced with permission from²⁵; Copyright 2022 Elsevier. Reproduced with permission from²⁶; Copyright 2023 Authors. Reproduced with permission from²⁷; Copyright 2024 Wiley-VCH. Reproduced with permission from²⁸; Copyright 2022, Author(s)).

wires in the 12-lead system limit the patient's range of motion, causing considerable inconvenience. In view of these problems, it is particularly important to explore dry electrode techniques that do not rely on wet gels, combined with simplified lead configurations to obtain accurate ECG signals.¹¹

Dry electrodes are categorized into two types according to their flexibility: nonflexible dry electrodes and flexible dry electrodes. The former are mostly made of metals such as copper (Cu), Ag, gold (Au), ceramics, and silicone (Si). Although they have excellent electrical conductivity, direct contact with the skin limits their use in long-term ECG monitoring.¹² The latter are usually made of flexible materials such as stretchable elastomers or hydrogels composited with conductive materials, combining high conductivity and high flexibility, making them virtually immune to the sensation of foreign bodies when worn. For long-term ECG monitoring, among flexible dry electrodes, e-skin/tattoo electrode, textile-based electrode, and microarray electrode are usually used due to their unique flexibility and easy-to-integrate characteristics. More importantly, they are capable of detecting signals stably when the body is in motion, which makes them ideal for use in long-term and dynamic ECG monitoring. Therefore, flexible dry electrodes (FDEs) are regarded as an ideal choice for the next generation in the field of ECG monitoring.^{11,13}

In the field of flexible wearable health monitoring, most of the articles reviewed mention advances related to ECG monitoring

technology, but reviews addressing advances in ECG electrodes are lacking.^{14–16} However, due to the development of various novel technologies, the emergence of various highly integrated devices, and the continuous expansion

of application areas, it is necessary to continue reporting the latest advances. In this study, we report the basic working principles, electrode design strategies, and recent advances in advanced electrodes and summarize the development and challenges of flexible cardiac electrodes (Figure 1). Chapter 2 describes the number of electrodes, the connection methods, the basic working principles, and the advantages and disadvantages according to the number of leads, from small to large. Chapter 3 describes the current design strategies for FDEs for ECG monitoring, including conductive materials, flexible stretching materials, and electrode structures. Chapter 4 introduces more cutting-edge FDEs for ECG detection, which mainly include three different types: electronic skin/electronic tattoo electrodes, textile-based electrodes, and microarray electrodes. Chapter 5 summarizes the current general problems of electrodes and provides possible directions for development. We believe that this review will provide some reference for the future development of ECG electrodes and spawn new opportunities.

ELECTRODE LEADS AND WORKING PRINCIPLES FOR ELECTROCARDIOGRAM

The accuracy and reliability of an ECG, a key tool for capturing the electrical activity of the heart, are highly dependent on the design of the ECG electrode lead (EEL) system. Traditional EEL systems, including but not limited to the standard 12-lead

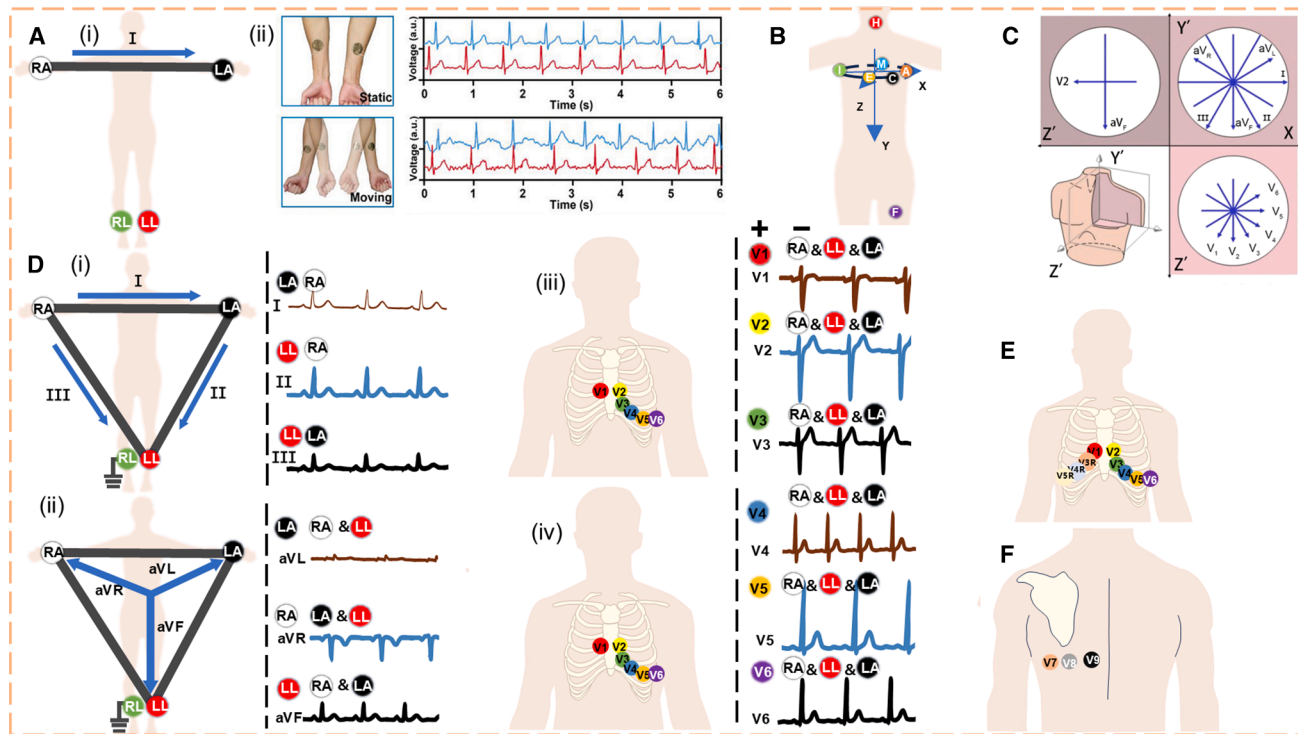


Figure 2. Electrode positions in different leads

(A) (i) Single lead electrode position for monitoring lead I; and (ii) Monitoring of single-lead electrodes in lead I and corresponding ECG signals. Reproduced with permission from ref.²⁹; Copyright 2020, WILEY-VCH.

(B) Frank orthogonal lead electrode positions.

(C) Twelve-lead projections in different cross-sections. Reproduced with permission from ref.³⁰; Copyright 2016, Elsevier Inc.

(D) Twelve-lead electrode positions and corresponding lead signals for (i) standard limb leads I, II, and III; (ii) pressurized unipolar limb leads aVR, aVL, and aVF; (iii) thoracic leads V1, V2, and V3; and (iv) thoracic leads V4, V5, and V6.

(E) Fifteen-lead (also known as the right thoracic lead) V3R, V4R, and V5R.

(F) 18-lead (also known as posterior wall leads) V7, V8, and V9.

system, provide us with a comprehensive view of cardiac electrophysiological activity (Figure 2). However, with advances in flexible electronics and smart wearable technologies, the design and application of EEL is undergoing a revolution. From simple monitoring with a single lead to comprehensive analysis with a multi-lead system, each lead configuration has unique advantages and limitations; thus, understanding these differences is essential for choosing the right ECG monitoring solution.

This section first describes single-lead systems, which provide basic ECG information with a minimal number of electrodes and are well suited for simple and portable monitoring applications. The 3-lead, 12-lead, 15-lead, and 18-lead systems are discussed in turn, each of which reveals different aspects of the heart's electrical activity through a specific electrode layout. In addition, this section will evaluate the performance of these lead systems in real-world applications, including their accuracy, patient comfort, and applicability in different health settings.

Single-lead electrocardiogram electrodes

Single-lead ECGs capture ECG signals using only a pair of specifically oriented electrodes (which may include a reference elec-

trode), a configuration that primarily reflects the electrical activity of the heart along a single axis. In practice, single-lead electrodes are usually placed at the wrist or chest (Figure 2A), for example, leads from the left wrist to the right one are used to monitor the electrical activity of the heart.^{31–33} The advantage of single-lead systems is their simplified design, which reduces the complexity associated with wires and makes ECG monitoring more convenient. There are a variety of these monitoring devices currently on the market, including bracelets, watches, chest straps, undershirts, and stick-on monitors. Notably, although these devices are categorized as single-lead systems, in some cases, such as bracelets and watches, they use photoelectric volumetric pulse tracing to indirectly measure heart rate and oxygen saturation, rather than directly recording ECG signals.^{34,35} Other devices offer truly single-lead ECG monitoring capabilities, allowing the user to obtain ECG data through simple manipulation. However, these systems have limitations in their diagnostic capabilities; they are unable to assess the electrophysiologic status of the heart as comprehensively as multi-lead systems. In particular, it is difficult to accurately identify ST-segment changes and types of arrhythmias because single-lead ECGs do not provide the precise spatial orientation

information of P-, QRS-, and T-waves, as well as detailed ST-segment dynamics.^{34,36,37}

Despite the limitations of single-lead systems in terms of their diagnostic capabilities, their portability, ease of operation, and potential for use in long-term monitoring in combination with multiple types of electrodes make them promising areas of research in the field of ECG monitoring. Researchers are actively exploring the application of advanced electrode technologies to single-lead ECG systems to improve their performance and clinical application value. For example, Wang et al.³⁸ prepared spongy aerogel electrodes (silver nanowires (AgNWs)/polypyrrole (PPy)-cellulose acetate (CA) with excellent flexibility and conductivity via thermally induced phase separation technology. This electrode was able to maintain a clear waveform during continuous monitoring for 24 h, demonstrating excellent long-term monitoring capability. Similarly, Zhao et al.³⁹ successfully prepared an ultrathin, highly electrochemically dry epidermal electrode by coating a poly (3, 4-ethylenedioxythiophene):poly (styrenesulfonate) (PEDOT: PSS) film on graphene grown by chemical vapor deposition (CVD), which demonstrated high signal-to-noise ratio (21 ± 0.3 dB) stability in dynamic monitoring. These studies suggest that by combining appropriate electrode materials and designs, single-lead ECG monitoring devices can achieve stable and reliable long-term ambulatory monitoring, which has broad application prospects.

3-Lead electrocardiogram electrodes

Compared with single-lead systems, 3-lead ECG monitoring systems provide a more comprehensive view of the electrical activity of the heart by adding up to 3 leads (Figures 2B and 2C). A classic example is the Frank orthogonal lead system proposed by Ernset Frank in 1956, which obtains 3D electrophysiological information of the heart through a carefully designed electrode layout.⁴⁰ As shown in Figure 2B, the system consists of 7 electrodes, including 5 chest electrodes (A, C, E, I, and M), a neck electrode (H), and a standard limb electrode (F) placed on the left leg. The exact placement of these electrodes is precisely planned to ensure that different aspects of the heart's electrical activity can be accurately measured. For example, the A and I electrodes form the X-lead, the F and H electrodes form the Y-lead, and the E and M electrodes form the Z-lead. The orthogonal alignment of these leads permits the capture of ECG signals from three mutually perpendicular directions, which in turn yields the electrophysiological data of the cardiac state. These data allow accurate diagnosis of ischaemia and acute myocardial infarction, providing important information for clinical diagnosis.⁴¹

In addition to the Frank orthogonal lead system, another 3-lead system is commonly used in ambulatory electrocardiography (AECG), which usually consists of 5 or 7 wires, including 3 bipolar leads CM1, CM2, and CM5, which simulate the V1, V2, and V5 leads in 12-lead; and a MaVF lead that simulates the aVF lead in 12-lead. This system was originally designed as a 7-lead configuration with each lead connected to a specific electrode, but to improve patient wearing comfort, the technical team reduced the number of wires to 5 by sharing the negative electrode.⁴² However, this electrode may affect signal clarity in the presence of electromagnetic interference, so the original 7-

wire configuration is still preferred. Among the 7-electrode systems, the CM1, CM5, and CMaVF leads are of significant clinical value for the diagnosis of cardiac conditions such as arrhythmias, myocardial ischemia, coronary artery spasm, and variant angina pectoris.⁴³ Although the 3-lead system provides valuable information in ECG monitoring, it has limitations in detecting ST-segment downshifts and pinpointing arrhythmias because of the limited number of leads.⁴⁴ Therefore, this type of monitoring system is more suitable for patients with health checkups, mild symptoms, or patients who cannot be monitored via conventional multi-lead monitoring due to special circumstances (e.g., postoperative trauma, large skin burns). In addition, it is also adaptive for situations where rapid or preliminary ECG assessment is needed.

12-Lead electrode electrocardiogram

A standardized 12-lead system for ECGs, consisting of 10 electrodes, can provide 12 different views of the ECG signal (Figure 2D(i–iv)). This system, originally proposed by Willem Einthoven, consists of four electrodes for the extremities (left leg LL, right leg RL, left arm LA, and right arm RA) and six chest electrodes (V1 to V6). The configuration of the extremity electrodes forms standard limb leads I, II, and III, and the pressurized limb leads aVR, aVL, and aVF. Moreover, thoracic electrodes are used to form the thoracic leads V1 to V6. The precise location and correspondence of these leads are critical for ECG interpretation.

The positive and negative poles of each lead can be visualized as an imaginary connecting line called the lead axis. The 12-lead system consists of a frontal six-axis system and a transverse six-axis system, where the frontal one consists of leads I, II, III, aVR, aVL, and aVF, whereas the transverse one consists of leads V1 through V6. The depolarization and repolarization processes of the heart produce a 3D ECG vector loop. The projections of this loop on the frontal and transverse planes form the frontal-plane vector loop and the transverse-plane vector loop, respectively. The projection of the former onto the frontal-plane six-axis system produces ECG waveforms in leads I, II, III, aVR, aVL, and aVF, whereas the projection of the latter onto the transverse-plane six-axis system produces ECG waveforms in leads V1 through V6. The former provides feedback on the electrical activity of the heart in the anterior–posterior, left–right, and up–down directions, and the latter reveals the electrical activity of the heart in the anterior–posterior and left–right directions. By combining and analyzing data from multiple leads of ECGs, physicians can accurately diagnose a patient's cardiovascular health. For example, lower wall myocardial infarction can be recognized in leads II, III, and aVF, whereas leads V1 through V6 help detect anterior wall myocardial infarction.^{45,46} However, the conventional 12-lead system of ECG suffers from monitoring blind spots, particularly in the right ventricle and posterior wall of cardiac activity, areas that are often not visible on a conventional ECG.

In standard 12-lead ECG monitoring, Ag/AgCl-hydrogel electrodes are often used to improve signal quality or as control electrodes to test signal quality.⁴⁷ The hydrogel serves to increase the contact area between the electrode and the skin, decreasing impedance and resulting in a clearer ECG signal. However, these

electrodes may encounter problems after prolonged use because the water in the hydrogel gradually evaporates, resulting in an increase in contact impedance and a decrease in signal quality; in addition, prolonged contact with the hydrogel may cause allergic reactions, such as redness and swelling of the skin. To solve these problems, researchers have developed novel hydrogel materials, such as PEDOT:PSS polymerized in polyvinyl chloride and carboxymethyl cellulose matrices developed by Wang et al.,⁴⁸ which retain moisture and maintain good ECG signal quality even after 28 days. Xiao et al.⁴⁹ reported that a poly (vinyl alcohol) (PVA)/polyethylene glycol (PEG)/graphene oxide GO conductive gel prepared by cyclic freezing and thawing can be used in conjunction with Ag/AgCl hydrogel after 6 h of wear, in addition, it can reduce the occurrence of allergy and itching. These improvements demonstrate that the comfort and accuracy of ECG monitoring can be significantly improved with innovative materials. Nonetheless, the multiple wires involved in the 12-lead system may still be inconvenient for patients, especially if prolonged or dynamic monitoring is needed.

15-Lead and 18-lead electrode electrocardiogram

To compensate for the inadequacy of 12-lead ECGs in monitoring right ventricular and posterior wall myocardial infarction, the medical community developed 15-lead and 18-lead ECGs (Figures 2E and 2F). The 15-lead complemented the monitoring of the right ventricle by adding leads V3R, V4R, and V5R in symmetrical positions on the right chest, whereas the 18-lead further added leads V7, V8, and V9 to the 15-lead to cover the posterior wall of the left ventricle. Unlike the 12-lead, which only covers the anterior and lateral walls of the heart, the addition of V3R-V5R and V7-V9 provides supplementary cardiac information regarding the posterior wall and posterior wall of the right ventricle. This reduces misdiagnosis, such as acute myocardial infarction caused by diagnostic blind spots affecting the posterior wall and posterior ventricles. These extended leads significantly improve the ability to monitor overall cardiac activity, specifically in recognizing premature ventricular contractions, myocardial infarction, and arrhythmias, showing significant advantages over 12 leads.^{50–53} However, while 15-lead and 18-lead methods provide a more comprehensive view, they are not uncontroversial. By further utilizing lead conversion techniques to create a mathematical model of the chest leads of a normal ECG and comparing it to a rotational ECG model, the researchers reported that the right chest lead and the posterior wall lead were, to some extent, reflections of the left chest and forearm leads, suggesting an overlap of information. In addition, the multiplicity of electrodes and wires poses challenges in terms of portability and ambulatory monitoring, as the complex setup process and the requirement for a patient's resting state limit the convenience of these leads in practical applications.

ELECTRODE DESIGN STRATEGIES

As the basis of cardiac electrodes, conductivity determines the advantages and disadvantages of electrodes; correspondingly, the selection of electrode materials determines the level of conductivity, and it is obvious that electrode materials play a decisive and key role in electrodes. In addition, the various structures

of the electrode can give the electrode other functional characteristics (breathability, flexibility, and so forth) in addition to conductivity; therefore, the electrode structure also has a significant effect on the electrode. In this section, the conductive materials and electrode structures of flexible ECG electrodes are introduced and analyzed, and related electrode design strategies are given.

Conductive materials

In general, ECG electrodes require excellent electrical conductivity to collect high-quality ECG signals, and the importance of conductive materials is obvious as the first link in electrode design. Currently, common conductive materials for electrodes can be categorized into carbon-based materials,^{54,55} metal-based materials,^{56,57} and conductive polymers⁵⁸ (Figure 3). By skillfully combining these materials, the overall conductivity of the electrodes can be significantly improved, thus ensuring that the captured ECG signals are of high fidelity and clinical value.

Carbon-based conductive materials are mainly different-dimensional nanometre-scale carbon materials (Figure 3A), ranging from early carbon black (CB)⁶⁶ to present-day materials in several categories (0D CB,⁵⁹ 1D carbon nanotubes (CNTs),⁶⁰ 2D graphene,^{67–70} and so forth). Owing to the formation of π -bonds between carbon atoms, π -electrons are highly delocalized in the honeycomb lattice, forming a free-moving electron cloud, which allows the electrons to move freely or jump between them, thus conferring a high electrical conductivity (10^3 S/cm $\sim$ 10^7 S/cm, four-point probe) to the carbon-based materials.⁷¹ In addition, carbon nanomaterials generally have high flexibility and high mechanical properties (tensile strength $\sim$ 10–60 GPa and Young's modulus up to $\sim$ 1 TPa, universal testing machine tensile testing), which synergize to make them promising for various applications.⁷¹ However, they are prone to aggregation due to intermolecular van der Waals forces, which has become a common drawback.⁷² In addition to the use of commercial dispersants,⁷³ researchers have developed new multistage dispersants^{22,23} to synchronize the dispersion of various of these materials, the use of oxygen-containing groups (polyether amines, and so forth), and the mutual attraction or repulsion of hydrophobic groups. These new dispersants have achieved excellent dispersion effects and do not affect electrode flexibility, mechanical properties, or other characteristics, which are promising for application. In addition, using special technology^{74,75} to assist in dispersion is also a good way. Notably, the dosage of dispersants needs to be accurately controlled to avoid excessive micellar behavior, which may prevent the formation of a continuous conductive medium. Furthermore, the low electronic conductivity,¹⁰ complicated synthesis process, unfriendly environment, and unsatisfactory mechanical properties of electrodes that exist in the carbon-based part of the material can be improved by imitating the structure of nature,^{76,77} introducing special processes,^{76,78,79} and compositing with other materials,^{22,47,80,81} so that electrocardiographic electrodes with high electrical conductivity can be obtained with high flexibility and strength at the same time.

Metallic materials with stable ultrahigh electrical conductivity ($\sim$ 10^5 S/cm, four-point probe)⁸² are widely used for the

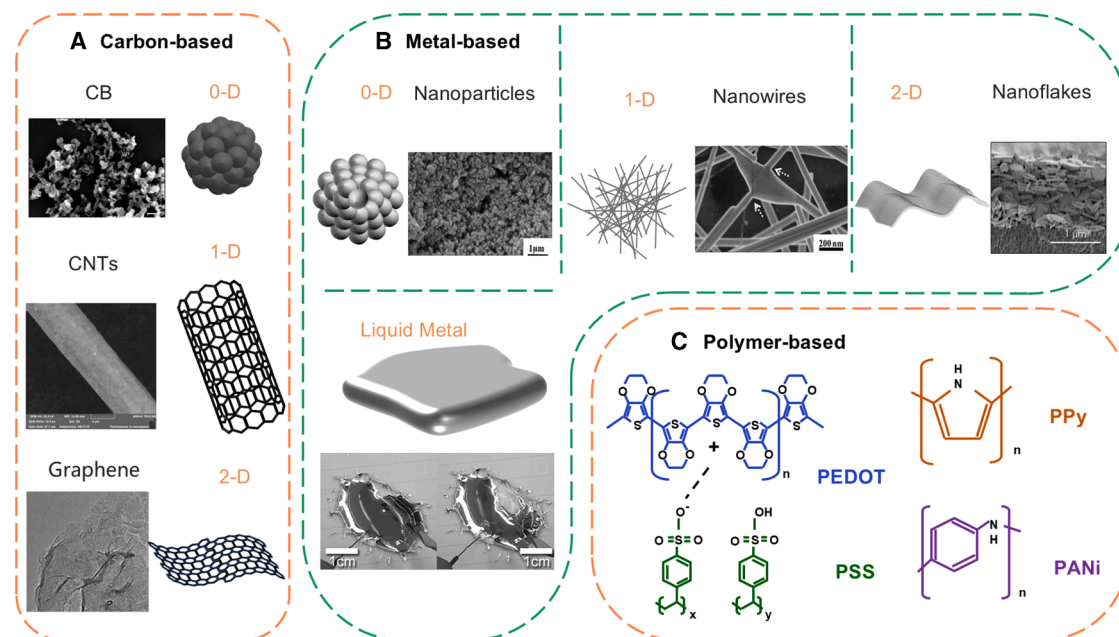


Figure 3. Different kinds of electrode conducting materials

(A) Carbon-based conductive materials, including zero-dimensional CB (top left and top right), reproduced with permission from⁵⁹; Copyright 2015, American Chemical Society; one-dimensional CNTs (middle left and middle right), reproduced with permission from⁶⁰; Copyright 2016, Elsevier Ltd.; and two-dimensional graphene (bottom left and bottom right), reproduced with permission from⁶¹; Copyright 2009 WILEY-VCH; respectively.

(B) Metal-based conductive materials, which are divided into two categories: metal nano-conductive materials (top), reproduced with permission from⁶²; Copyright 2014 WILEY-VCH; and liquid metals (bottom), where metal nano-conductive materials are subdivided into zero-dimensional metal nanoparticles (top-left), reproduced with permission from⁶³; Copyright 2018, Author(s); one-dimensional metal nanowires (top-center), reproduced with permission from⁶⁴; Copyright 2014 WILEY-VCH; and two-dimensional metal nanoflakes (top-right), reproduced with permission from⁶⁵; Copyright 2022, Authors.

(C) Conducting polymers, including polythiophene, with PEDOT:PSS as an example, PPy and PANI.

fabrication of flexible dry electrodes (Figure 3B). Furthermore, novel types of metallic nanomaterials and liquid metals,^{62,63} among others, also combine good mechanical flexibility. The former is similar to carbon materials and includes 0D metal nanoparticles,⁸³ 1D metal nanowires,⁶⁴ and 2D metal nanoflakes,⁶⁵ which can be synthesized by electrochemical deposition, chemical reduction, and so forth, and used for the fabrication of electrodes. Among all three materials, 1D metal nanowires have gained attention because of their ultrahigh aspect ratio and ability to easily form permeable networks.^{84,85} Generally, these represents a higher conductivity of the permeation network, which is due to the presence of fewer contact sites in the disordered alignment of the material, which reduces the electrical resistance; moreover, the special aspect ratio also implies excellent stretchability compared with that of nanoparticles and nanoflakes, and thus, attention should be focused on the selection of them in the process of fabricating the electrodes.^{56,57} In addition, other conditions during the fabrication of electrodes, such as deposition methods and postprocessing methods, should also be emphasized to obtain higher electrical conductivity.^{86–88} The latter, liquid metal alloys, are considered promising materials for the formation of high-performance flexible electrodes because of their intrinsically high fluidity and high electrical conductivity.^{89,90} Electrodes that use them as conductive materials, such as fibers and textiles, can also achieve very high strains (~1,500%, universal testing machine tensile testing) while

ensuring high electrode signals^{91–93} and can also be fabricated into conductive inks to form electrodes at a low cost. However, since they are prone to oxidation, it is important to focus on sealing the electrodes during electrode fabrication.⁹⁴

Some of the conjugated polymers in polymer materials can transport electrons by forming a continuous π -electron cloud of carbon atoms, which, together with the mechanical flexibility caused by polymer chain dynamics, makes these materials usable to produce flexible electrodes (Figure 3C). The common PEDOT:PSS,^{95,96} polyaniline,^{97,98} and PPy^{99,100} belong to this type of material. In addition, poly(3-hexylthiophene)¹⁰¹ and polymers based on the diketone pyrrolylpyrrole¹⁰² can be used to make semiconductors. These conductive materials rely on high crystallinity to maintain high conductivity; however, they can be damaged by external forces, so methods such as fine-tuning the internal polymer morphology or blending with elastomeric polymers are used in the fabrication of electrodes to ensure high conductivity and mechanical properties.¹⁰³ Some of them have unique intrinsic stretchability and can be used as both conductive and stretchable materials. However, these materials typically have low conductivity, and their unique amorphous structure allows them to obtain some stretchability and limits the crystallinity, further limiting the stretchability of the material. In addition, a few materials have low wet stability, which can be addressed by compositing them with other types of materials.^{104–108}

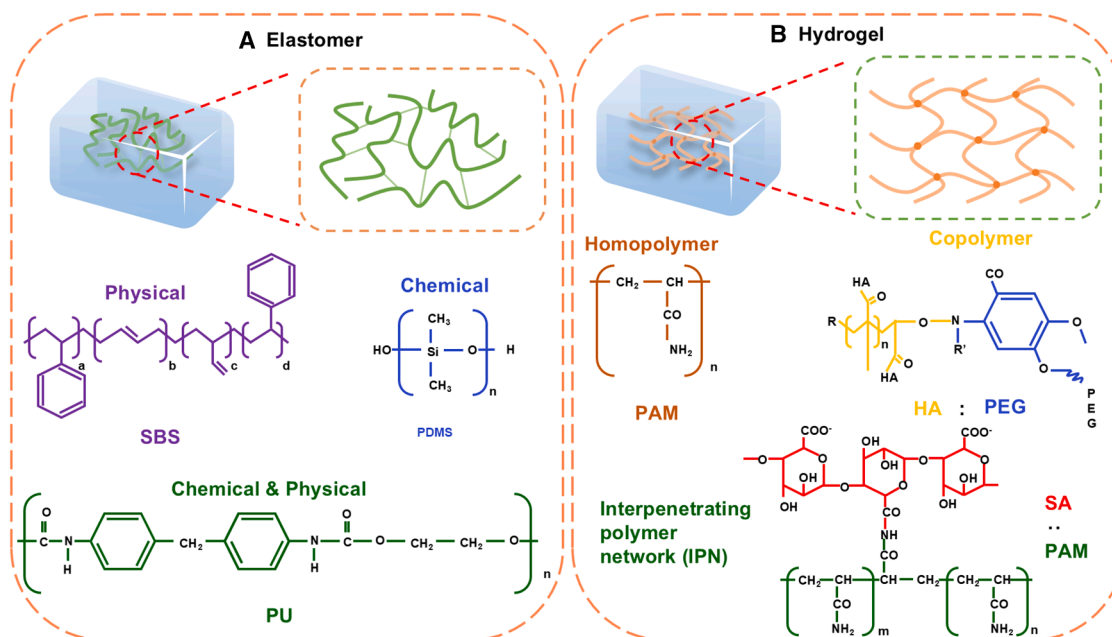


Figure 4. Flexible matrix material

(A and B) The left side of Panel (A) shows the elastomers, the left side of the middle shows the physically crosslinked mold SBS block copolymer, the right side of the middle shows the chemically crosslinked molded PDMS, and the bottom side shows the physically and chemically crosslinked comold PU. The right Panel (B) shows a hydrogel with homopolymerized PAM on the left side of the middle and copolymerized HA: PEG on the right side of the middle. And SA: PAM with an interpenetrating binetwork on the lower side.

Flexible matrix materials

Conductivity is the foundation of the electrode, which fundamentally determines the electrode's ability to collect and conduct signals; mechanical properties (Young's modulus, tensile strength, and so forth) are the electrode's progression, which centrally decides the electrode's flexibility and comfort. Therefore, in the design of the electrode, the flexible matrix material (Figure 4) is an indispensable second link, which, together with the conductive one, plays a decisive role in the electrode.

Elastomers are preferred as flexible matrix materials for electrodes since they can form a network by cross-linking and protect the permeating conductive material by dissipating the applied strain, thus achieving good mechanical properties (Young's modulus for up to ~ 105 kPa) (Figure 4A). Specifically, elastomers form side chains between neighboring polymer chains during cross-linking, resulting in a web-like interconnected network that prevents the individual slipping of polymer chains.^{84,109} The crosslinking of elastomers is divided into chemical crosslinking and physical crosslinking; the former forms a polymer network through strongly interacting covalent bonds. For example, polydimethylsiloxane (PDMS), Ecoflex, and silicone rubber materials are typical examples of chemical crosslinking.^{110,111} They usually react in the presence of a cross-linking agent, and by adjusting the agent's proportion or the reaction temperature, the density of chemical bonds can be altered to control the overall properties. These elastomers are quite stable in terms of their mechanical properties, but the amount of electrically conductive material in the elastomer will generate a different number of cross-linking points, which will

change the overall flexibility.¹¹² In addition, this type of network may cause severe elastomer swelling when exposed to organic solvents.⁸⁴ The latter cross-linking occurs, for example, through hydrogen bonding, which is weaker than covalent bonding. Typical examples of such materials are styrene-based elastomers, including poly(styrene-butadiene-styrene) (SBS) or poly(styrene-ethylene-butylene-styrene) (SEBS), which consist of alternating soft segments that provide flexibility and hard segments that provide stiffness; thus, the mechanical properties can be varied by controlling the ratio of soft to hard segments.²⁴ Unlike chemically crosslinked materials, the weaker bonding results in the inferior stability of the mechanical properties. Interestingly, this weakness also results in excellent processability, lowering the threshold for the application of physically crosslinked elastomers. In addition, physically crosslinked elastomers can be solidified and moulded even under conditions of high conductive material content. In addition to these two crosslinking methods, there are also physical and chemical crosslinking elastomers; the most typical example of such materials is polyurethane (PU)-based elastomers.¹¹³ The synthesis of these elastomers, represented by PU, is achieved by the reaction of large-sized polyols and di-/multi-isocyanates. Similar to SEBS/SBS, PU-based elastomers consist of alternating hard and soft segments, with the former formed by long chains of polyols and the latter formed by the reaction of isocyanates and short diols to obtain urea groups.¹¹⁴ In this type of elastomer network, the isocyanate groups are chemically crosslinked by sub-urethane bonds, and the urea groups are physically crosslinked by hydrogen bonds so that the overall mechanical properties

can be controlled by changing the type and ratio of the polyol and isocyanate. In addition to traditional elastomers, those with self-repairing or self-healing capabilities have gained widespread attention. These advanced elastomers can self-repair or self-heal by adding repair agents or introducing dynamic cross-linking. The former, through the use of reactive precursors and catalysts, synergistically leads to self-polymerization to repair damage; the latter utilizes the interactions between hydrogen bonding or metal-ligands and abundant dynamic bond sites to achieve self-healing properties.^{115–117}

Like elastomers, hydrogels are also physically or chemically crosslinked to form a 3D network structure, and have similar mechanical properties (Young's modulus up to ~ 104 kPa) and tissue-like structures, which have enabled the widespread attention to them in recent years (Figure 4B).¹⁰⁹ Some of them are similar to elastomers, and the 3D network structure is formed by the polymer long chain and its side chains, which are interlocked to form a framework that is less prone to slip; others rely on electrostatic forces similar to those of positive and negative ions to form a 3D network structure, which easily slips and damaged by the impact of the external force on the long chain; moreover, they will swell while in the wet state, and their mechanical properties will be impaired. There are three types of network structures of hydrogels: homo-polymeric, copolymeric, and interpenetrating networks. The first types are synthesized from a single type of monomer, such as polyacrylamide (PAAM), polyacrylic acid (PAA), or PEG, which are formed by a single cross-linking agent; thus, a simple change in the agent content or cross-linking conditions can control the flexibility of the hydrogel but results in a low fracture energy (<10 J/m², in contrast to ~ 2.5 kJ/m² for skeletal muscle, ~ 1.5 kJ/m² for skeletal muscle) and difficulty in balancing various mechanical properties. An increase in the elastic modulus/strength often implies a decrease in the tensile/fatigue threshold.¹¹⁸ The second types are formed by the copolymerization of two or more components and is moulded by one or more steps of cross-linking. PEG-hyaluronic acid (HA) copolymerized hydrogels are representative of this type, which includes microscopic phase-separated structures in the network, where the PEG forms a continuous phase that provides a stable network structure and the HA forms a dispersed phase; together, they can ensure high modulus; thus, the mechanical properties can be adjusted by controlling the ratio between them.¹¹⁹ This design strategy is more versatile, but it is still difficult to balance their strength and flexibility. Interpenetrating hydrogels contain multiple independent gel networks that can fully utilize their properties. The double-network polyacrylamide-sodium alginate (SA) hydrogel is a typical example; the propylene double bond of the former and N, N'-methylene bisacrylamide (MBAAM) generate free-radical polymerization, and the latter's guluronic acid (G unit) generates electrostatic adsorption with the ions so that the control of both the content of cross-linking points and the cross-linking conditions can control the mechanical properties.¹²⁰ More importantly, this network structure can equalize these properties and thus has good potential for development. Recent hydrogel designs have focused on achieving elastomeric mechanical properties by mimicking the multilayered cross-linking biostructures of spiders/introducing special energy dissipation mechanisms,

but this type of design lacks universality and is difficult to apply on a wide scale.^{121,122} In summary, hydrogels mechanics are affected mainly by the uneven distribution of polymer density/uneven polymer chain length/lack of energy dissipation mechanism/high water content. Therefore, their design with various excellent mechanical properties should integrate four aspects, such as the topological design of the slip ring structure/the use of bond sacrifices to introduce energy dissipation/the introduction of a high-order structure, and other methods to improve the mechanical properties. For more information about the design of hydrogels with high mechanical properties, several reviews can be found in the literature.^{118,123}

Electrode structure

The electrode materials determine the strength of the basic performance of the electrode, whereas its structure (Figure 5) decides the quality of the target performance. Therefore, the electrode structure design is the third link after the electrode conductive material and the flexible tensile material, which distinguishes the electrode, as a whole, as another major component of the electrode design strategy.

Nano structure

A previous detailed description of materials revealed that carbon-based and metal-based nanomaterials play crucial roles in the field of flexible electrodes because of their excellent conductive properties; similarly, nanoscale structures that can be introduced in the design of electrode structures are equally important. The so-called nanostructures are those precisely constructed on the scale of 1–100 nm. As shown in Figure 5A, nano-meshes¹²⁹ (Figure 5A(i)), nanoholes^{126,130} (Figure 5A(ii)), and nanofibers (Figure 5A(iii)) belong to this structure. Their unique physical phenomena, such as large specific surface areas, size confinement effects, and dielectric properties, greatly enhance the electron transport capacity and interfacial activity of materials, thus establishing the dominance of nanostructures in the design of flexible electrodes. Their fabrication processes can be categorized into two main types: top-down/bottom-up methods. The former transforms large materials into surface structures with nanoscale features through fine etching techniques such as photolithography and e-beam lithography. In contrast, the latter relies on self-assembly at the molecular or atomic scale, where basic building blocks are progressively stacked to form complex macroscopic structures via techniques such as CVD, magnetron sputtering, and electroplating. Each of these methods has its own advantages and offers a diverse range of pathways for realizing high-performance nanostructures. Given the simplicity of the fabrication process and the excellent overall performance, nanofiber structures prepared via advanced processes such as electrostatic spinning¹³¹ and microfluidic spinning¹³² have been widely adopted in several applications of FDEs. For example, Qiu et al.⁷⁶ successfully constructed nano-fibres with bird's nest-like microstructures via the first technology, which significantly improved the physical and mechanical properties of graphene thin-film electrodes and ensured the washability and reusability of the electrodes. Li et al.¹²⁵ integrated a nanofiber structure with CNTs arrays cultivated by CVD and provided an efficient path for electron

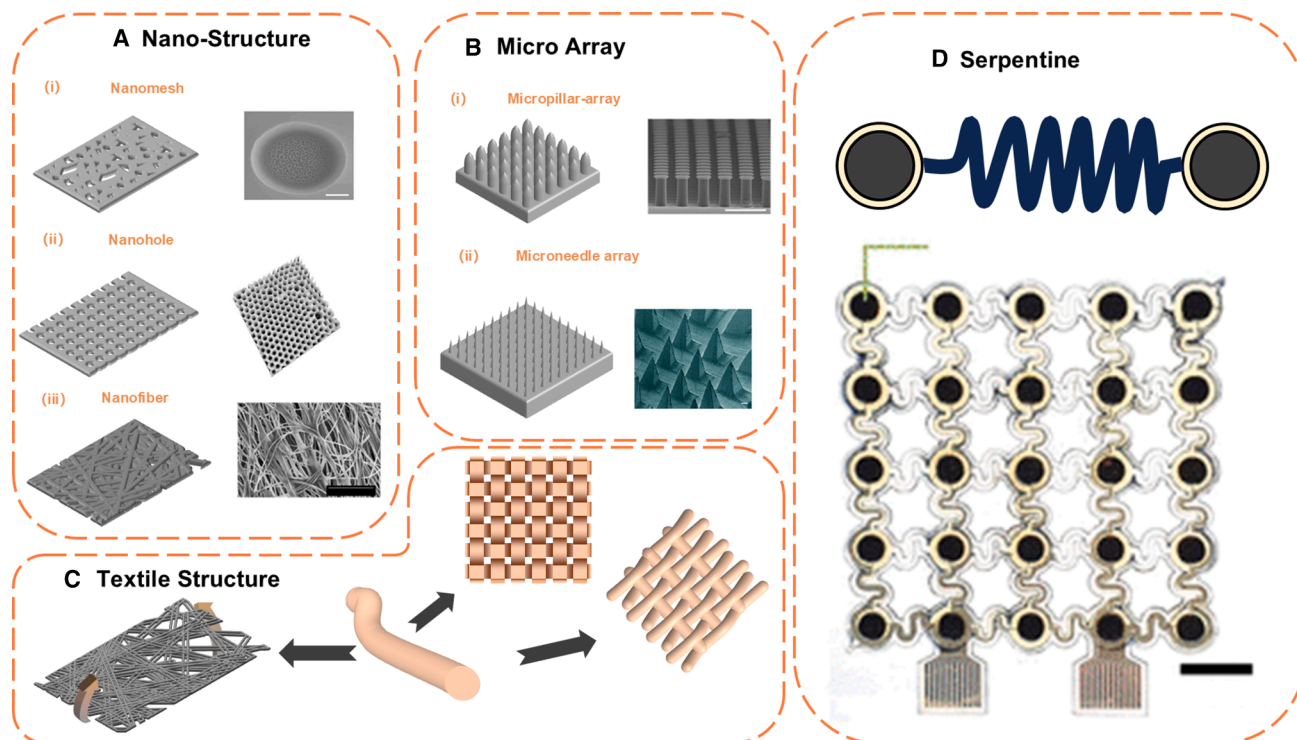


Figure 5. Electrode structure

(A–D) The left side of the upper (A) shows nanostructures, where the upper part (i) is the nanomesh, reproduced with permission from¹²⁴; Copyright 2024 Wiley-VCH; the middle part (ii) is the nanopore, reproduced with permission from¹²⁵; Copyright 2020 Elsevier Ltd.; and the lower part (iii) is the nanofiber, reproduced with permission from¹²⁶; Copyright 2017 WILEY-VCH. The middle side of the upper (B) is the microarray structure, where upper (i) is the micropillar-array, reproduced with permission from⁵⁴; Copyright 2016 American Chemical Society; and the lower part (ii) is the microneedle array, reproduced with permission from¹²⁷; Copyright 2020 WILEY-VCH. The left side of the lower (C) is the textile structure, the left part is the base of the fiber membrane, the middle side is the single fiber after curling, and the right sides are the fabrics that can be obtained after spinning and other methods. The right side (D) is the serpentine structure, reproduced with permission from¹²⁸; Copyright 2024 The Authors.

transport through its 45° angled diagonal arrangement, thus realizing the high signal acquisition quality of thin-film electrodes. Yang et al.²¹ prepared an all-nanofiber structure electrode using the same technology, which is capable of spontaneously removing body fluids and preventing their reflux by virtue of its unique structural design. In addition, nanomesh structures and nanoholes are valued for their excellent air permeability; the former further enhances the sensor performance by increasing the conductive pathways, whereas the latter is favor for its contribution to electrode sensitivity.^{133–135} Lee et al.¹³⁶ developed cardiac electrodes using a PU nanofiber mesh, which demonstrated excellent air permeability and high conductivity. Ma et al.¹³⁷ optimized the fabrication process of nanohole structures via the emulsion template method to prepare cardiac electrodes with high sensitivity.

Array structure

Array structures, especially biomimetic micropillar structures, are high-tech products that mimic the microstructures of living organisms in nature (Figure 5B). Drawing inspiration from the fine arrangement of bird feathers and the microscopic filamentous hairs on the soles of gecko feet, they not only replicate the unique physicochemical properties of living organisms but

also demonstrate excellent performance in terms of adaptability and functionality, and thus have received extensive attention from the scientific research community.^{138–142} As shown in Figure 5B(i), biomimetic micropillar structures are highly favored due to their unique advantages, namely, high ordering, nano-scale effects, and excellent biocompatibility, which make them ideal for the development of advanced biomaterials with tailored functionalities and properties.^{143–146} The delicate design of such structures not only promotes favourable interactions between materials and biological tissues but also opens new possibilities for applications in fields such as biosensors, drug delivery systems, and regenerative medicine. By precisely controlling the geometry, size, and alignment of the micropillars, the behavior of cell attachment, growth, and tissue regeneration can be regulated, thus realizing the precise regulation of biological processes. Kim et al.⁵⁴ successfully constructed array electrodes with these structures via a compounding strategy of carbon nanofillers and PDMS, which showed excellent adhesion and signal acquisition ability in electrocardiogram (ECG) monitoring.

Microneedle array structures, with their precise needle-like features and tiny size (100~1,000 μm), play an important role in the bioelectronics field (Figure 5B(ii)). With tips and base diameters precisely regulated between tens μm and hundreds of

microns, they are usually applied to the skin in the form of micro patches, which take advantage of their ability to penetrate the stratum corneum to significantly reduce the contact impedance between the skin and the electrodes and efficiently minimize the signal artifacts induced by motion.^{33,54,147} Its fabrication process involves a variety of high-tech methods, including wet or dry photolithography, laser cutting, and micromachining technologies, which are capable of selectively sculpting the desired structure on a 2D substrate.²⁵ Moreover, innovative methods such as self-assembly, magnetorheological stretching drawing lithography (MRDL)^{33,148} have also been used to directly construct these arrays from tiny droplets or 2D planes, providing greater flexibility and innovation in the design of microneedle morphology and function. In recent years, microneedle array structured electrodes have become a research focus in bioelectronics. Ren et al.^{33,148} successfully fabricated them via MRDL, whereas Xia et al.²⁵ used conventional photolithography and wet etching to fabricate this type of electrode for ECG signal measurements on African clawed toads. Moreover, using the reactive ion etching (RIE) process, researchers have been able to develop ECG electrodes capable of long-term monitoring of cardiac activity in dynamically moving rats, which signifies the potential of microneedle array technology for application in bio-monitoring and diagnostic devices. Nonetheless, array structured electrodes still face several technical challenges in manufacturing and application practice. These include how to precisely regulate the micromorphology and arrangement pattern of the electrodes to optimize their overall performance; how to adjust and improve the interfacial interaction between the electrodes and the electrolytes in the organisms to increase the efficiency and long-term stability of their reactions; and how to effectively reduce the production cost to promote the large-scale array structured electrodes. Solving these problems could pave the way for the widespread application of flexible bio-electronic devices.

Textile structure

Textile structured electrodes represent a revolutionary integrated solution that skillfully blends traditional electrode functionality into a textile structure. As shown in Figure 5C, by weaving conductive fibers or yarns into textiles through woven, knitted, nonwoven, or braided techniques, it is possible to create FDEs that possess both conductive and sensing properties without losing the inherent softness, wearability, or comfort of the fabric. This innovative integration not only broadens the scope of application of electrodes but also greatly improves the wearer's experience, realizing the harmonious unity of functionality and comfort.^{80,149,150} For example, Le et al.¹⁵¹ optimized the roll-to-roll manufacturing process to produce yarns with uniform and stable Ag/AgCl coatings under the best fabrication process and obtained textile FDEs via an embroidery process. The electrode signals exhibited high resolution (higher r-peak amplitude, lower s-wave), low impedance (less than 2 M Ω between 0.1 and 150 Hz), and similar signal quality to standard electrodes (the standard deviation and coefficient of variation were almost the same), and could be used for ECG signal acquisition. Hossain et al.⁸¹ noted that the bonding of electrodes to the skin by van der Waals forces is prone to slip and fall off, so

the research group formed a conductive yarn with a skin-core structure by wrapping a layer of CNTs around ordinary yarns (cotton, spandex) with thermoplastic polyurethanes (TPU) and then weaving them to obtain CNT-based woven, knitted, and nonwoven textile electrodes. They are suitable for different scenarios, but they can rely on themselves to collect ECG data comparable (with similar signal-to-noise ratios) to those of commercial Ag/AgCl electrodes and can be worn and washed such as ordinary fabrics. Dou et al.¹⁵² reported that in conductive yarn, when stretched, the conductive layer is deformed. In addition, the inherent capacity of reduced Graphene Oxide (rGO) makes it more difficult for it to withstand multiple deformations of the conductive layer as the main material. Therefore, the group used friction spinning to obtain a textile structured electrode with a skin-core structure by compositing polyethylene terephthalate (PET) fibers coated with rGO/silver nanoparticles (AgNPs) with spandex. The electrode has good resistance to tensile deformation, high precision (5% difference from commercial electrodes), good bacteriostatic properties and sensitivity, and is suitable for ECG electrode monitoring. However, textile structured electrodes face some challenges in practical applications. The conductive layer prepared by techniques such as coating may be prone to peeling off under friction between yarns, which not only increases the contact impedance but also may reduce the quality of signal acquisition. In addition, compatibility issues between conductive materials and human skin may cause discomfort or foreign body sensation.^{153,154} To address these challenges, researchers have explored various improvement strategies, such as forming an AgCl layer on the electrode surface by electrochemical deposition or surface treating silver-plated fibers with nanoscale titanium (Ti) coatings.^{155,156} These innovative approaches not only enhance the adhesion of the conductive layer and reduce the risk of detachment but also significantly improve the biocompatibility of the electrodes, thus ensuring signal stability and reliability in long-term monitoring.

Serpentine structure

The design of serpentine electrodes (Figure 5D) plays a crucial role in FDEs, especially in applications that require conformal contact with human skin.^{157,158} The key to this design is to realize the stretchability and functionality of the device by controlling the interface between the rigid active material and the soft elastic material. Conventional rigid electronic devices are combined with soft connectivity through a serpentine bridge-island design strategy, where the "island" represents a rigid printed circuit board (PCB) based electronic device and the "bridge" represents stretchable serpentine conductive electrodes.¹⁵⁹ This design allows traditional non-stretchable medical devices to be interconnected with stretchable serpentine electrodes, greatly expanding their range of applications.¹⁶⁰ Serpentine electrodes typically consist of photolithographically patterned polymer films (e.g., polyimide (PI) and PET) and metal (e.g., Au and Cu) evaporations, which can be easily deformed to accommodate stretching, ensuring the functionality of the overall structure. The serpentine structure consists of both a 1D wave shape and a 2D network type. In the former case, the change in geometry when stretched corresponds to a change in amplitude to adapt

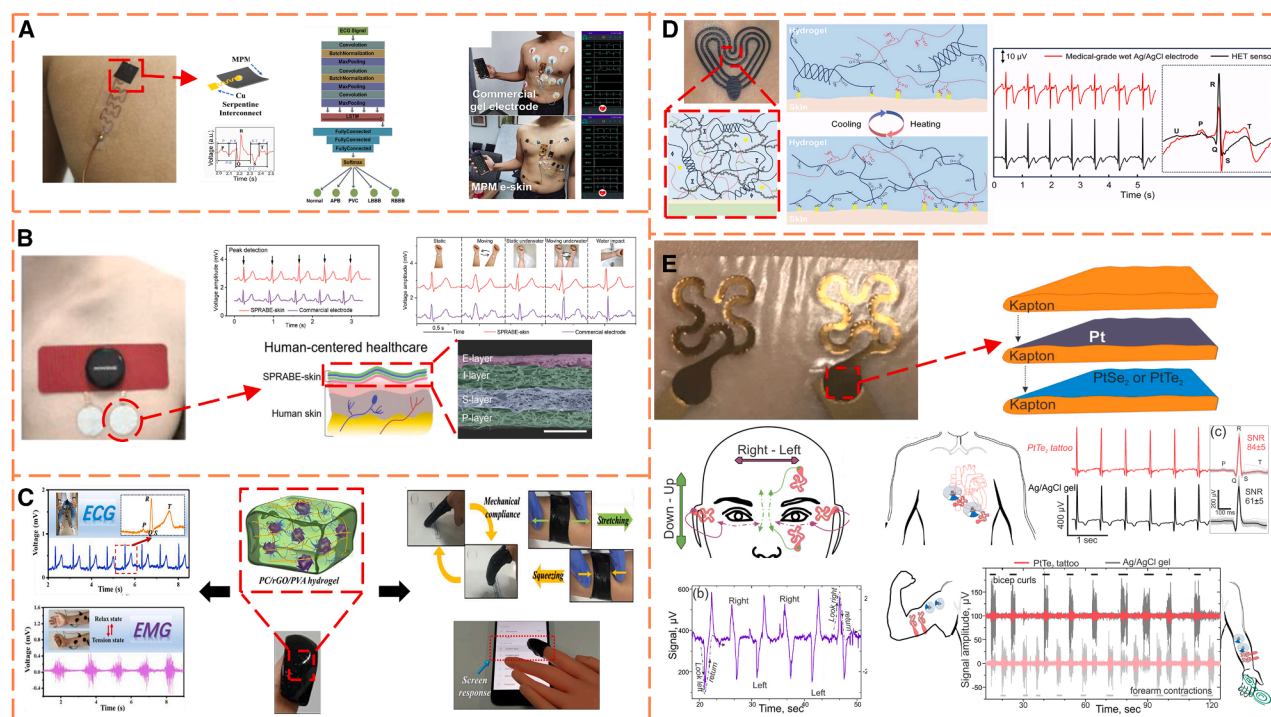


Figure 6. Advanced e-skin and e-tattoo ECG electrodes

(A) MXene-polyurethane mesh (MPM)-based e-skin, digital photo of e-skin, schematic diagram with its measured ECG data graph is on the left side, algorithmic model of the neural network is in the middle, and a picture of the linkage with a 12-lead ECG is on the right side. Reproduced with permission from²⁰; Copyright 2022 Elsevier (B). SPRABE-skin e-skin, of the digital photo is on the left side, and the ECG data in different cases with the e-skin schematic diagram is on the upper right side. Reproduced with permission from¹⁹; Copyright 2023 Wiley-VCH. (C) Composite hydrogel e-skin, with ECG and EEG data measured on the left, the hydrogel schematic and digital photo are in the middle, and e-skin multifunctionality, including supercompliance, superstretchability, and monitoring of biopower signals, is on the right. Reproduced with permission from¹⁶⁴; Copyright 2019 Elsevier. (D) Multilayer e-tattoo electrode, with a digital photo and schematic of the electrode is on the left, the recyclable properties of the hydrogel are in the center, and a graph of its measured ECG data is on the right. Reproduced with permission from¹⁸; Copyright 2024, Author(s). (E) Pt-TMD electronic tattoo, electrode digital photo, and schematic are on the top, electrode measured EEG, EMG and ECG image are on the bottom. Reproduced with permission from¹⁷; Copyright 2021 American Chemical Society.

to the strain rather than a stress concentration in the material itself, in which case out-of-plane deformation plays an important role in stretchability. The latter can be achieved by forming nanoscale connectivity networks through methods such as grain boundary lithography, and this high stretchability depends mainly on the elongated mesh in the elastic range and the distributed cracks outside the elastic range. The first type of connectivity can be easily deformed out-of-plane, whereas the second deforms the nanomesh into a larger network with a nested self-similar layout, which together result in a hierarchical network with large stretchability.¹⁶¹ A single waveform serpentine provides unidirectional stretchability, whereas the mesh serpentine adapts to strain in both directions. Further fractal design achieves the bidirectional stretchability of electronic materials by introducing fractal configurations such as Peano, Greek cross, and Vicsek, which form space-filling structures, and it has been shown that the stretchability of the connection can reach 308% when the fractal order is 4, which is significantly greater than that of a single waveform serpentine of approximately 100%.^{162,163}

ADVANCED FLEXIBLE DRY ELECTRODES FOR ELECTROCARDIOGRAM MONITORING

Electronic skin/tattoo electrodes

Inspired by human skin, electronic skin (e-skin) (Figures 6A–6C) has a promising future as a wearable electronic device, especially for FDEs, and is especially prominent in personal health monitoring.¹⁶⁵ Generally, it is made of easy-to-process materials, including liquid metals, nanoparticles, nanowires, and so forth, and has skin-like properties such as stretchability, breathability, and so forth.¹⁶⁶ The key to it is its unique signal collection and sensing capability, which can monitor various physiological signals, such as ECGs and electromyograms (EMGs), only by contact with the skin. Interestingly, it often contains a serpentine structure in which liquid metal and other materials act as a “bridge,” which not only ensures the signal transmission performance but also does not affect the electrode stretching permeability, effectively improving the applicability of the e-skin.¹²⁸ Cui et al.²⁰ developed an e-skin based on MXene-polyurethane mesh (MPM) and a corresponding system for the integrated

Table 1. FDEs and their conductivity, flexibility, comfort, signal quality and reusability

Electrode Type	FDEs	Conductivity	Young's modulus/ Conformability	Breathability/ Adhesiveness	Contact impedance/ SNR	Reusability
e-skin/tattoo	MPM e-skin ²⁰	–	Skin-conformable	2.1 kg/m ² /day; more than 7 days	4.68 k Ω at 1k Hz; 16.5 dB	30 min laundry
	HET ¹⁸	–	31 kPa	28 N/m (35°C)	234% lower at 20 Hz; 46.6 dB	10 times laundry
	Pt-TMD e-tattoo ¹⁷	1 k Ω /□	Skin-conformable	–	84 $\pm$ 6 dB	5 times laundry
Textile electrode	All-nanofiber-based electrode ²¹	4.3 Ω /sq	2.65 MPa	3753.1 kg/m ² /day; 43.087 N/m	24.6 dB	–
	Nanofiber membrane ²²	25 Ω /sq	Textile-based flexibility	8.5 kg/m ² /day; skin- adhesive	Lower than 50 k Ω at 10–1000 Hz	10 times laundry
	CNTT textile electrode ⁸⁰	6.6 $\pm$ 0.7 MS/m	Textile-based flexibility	Textile-like breathability	–	10 times laundry
Microarray electrode	LIG-based electrode ²⁶	30 Ω /sq	Conformal to forearm	Skin-adhesive	–	–
	SF-PU electrode ²⁷	–	Skin-conformable	–	9500–100 Ω at 1–100 Hz; 15.84 dB	–

realization of “monitoring-transmission-intelligent analysis” of wearable 12-lead ECGs (Figure 6A; Table 1). Specifically, the MPM e-skin is obtained by stripping the aluminum (Al) layer from the Titanium Aluminum Carbide MAX phase and laminating it with a PU mesh which is connected to the ECG cable by a serpentine copper wire, with MXene nanoflakes embedded in the electrodes and overlapping with those in the outer layer to form an inner and outer dual conductive layer, thus enabling the acquisition of ECG signals. These signals are subsequently processed through a flexible printed circuit board and transmitted to a hybrid algorithm model that combines a convolutional neural network and a short-term and long-term memory network for diagnosis, and the resulting accuracy of the relevant case signals is greater than 99%. In addition to their signaling capabilities, MPMs have excellent comfort, breathability, and high wash resistance, which makes them very promising for various applications. Notably, the accuracy of these signals is strongly affected by skin adhesion, and it is necessary to focus on improving adhesion. Hao et al.¹⁹ developed a novel biomimetic e-skin (SPRABE-skin) based on electrostatic spinning and spray-coating technology with the aim of developing an e-skin that can collect multiple biological signals (Figure 6B). The original P-S-I-E multilayer structure is the core of this e-skin. Specifically, the MXene-CNT@TPU membrane obtained by spray-coating technology is the strain-sensitive layer (S layer), which is responsible for collecting the strain signal and is flanked by a protective layer (P-layer) and an isolation layer (I-layer) composed of electrospun TPU membranes, which are responsible for resisting the effects of external changes and isolating the biomechanical/bioelectrical sensing layers to prevent mutual influences, respectively, and the MXene-WPU coated on the outside of the isolation layer serves as the electrode layer for bioelectrical potential sensing (E-layer). On the one hand, when the biopotential force is sensed, the conductive permeable network of MXene-CNT in the S-layer produces microcracks

when stretched, and elastic recovery is induced by the TPU network structure when released to achieve a force–electric response; on the other hand, the E-layer is in direct contact with the skin, thus utilizing ultralow impedance for the sensing of bioelectric signals. However, e-skin is sensitive to mechanical damage, and crack expansion after damage may lead to loss of permeability, degradation of the mechanical and electrical properties of the material, and even peeling and detachment. To solve these problems, researchers have fabricated self-healable e-skins via a variety of methods; however, it is still difficult for researchers to achieve an optimal balance between self-mechanical repair efficiency, electron transport efficiency, and mechanical properties, and such self-healable e-skins are still worthy of attention. Pan et al.¹⁶⁴ developed a novel neural network-like structure composite hydrogel e-skin and used this kind of electrode to obtain ECG signals comparable to those of commercial Ag/AgCl electrodes (Figure 6C). By connecting the proanthocyanidin/rGO/PVA hydrogel obtained via a two-step method with metal wires, the electrode combines the fast and highly self-healing properties of the hydrogel with the ultracompliance and ultrastretchability of the e-skin. It can monitor a variety of signals, including bioelectricity and biopower. For the former, the ECG signal PQRST wave captured by the e-skin is clearly recognizable, and the EMG electrical signal is highly accurate; for the latter, the electrodes can monitor a variety of situations, such as finger bending and changes in facial expression, which broadens the range of applications of this e-skin. Electronic tattoos (e-tattoos) (Figures 6D and 6E), as sophisticated sensing devices within the e-skin domain, focus on realizing highly conformable interfaces with the skin to significantly reduce the discomfort during wearing.¹⁶⁷ In recent years, significant progress has been made in this field, especially in enhancing skin compliance. While the microscopic roughness of the skin surface and the presence of hair limit the conformability between traditional thin and soft wearable electronic devices and the

skin, the design of e-tattoos overcomes these limitations. Their ability to fit seamlessly to the skin, independent of skin deformation, effectively inhibits shedding and relative movement at the skin–electrode interface due to motion, significantly reducing motion artifacts while maintaining the natural functionality and tactile feel of the skin. This innovative e-tattoo technology not only improves the reliability of the signal but also provides the possibility of long-term monitoring of wearable devices, opening new horizons for the development of e-skin technology.^{168–172} Current e-tattoos are made of conductive films whose skin–electrode interface dryness limits the complete fitting of the two, resulting in high impedance and high motion sensitivity at the interface. Based on this situation, Zhuo et al.¹⁸ developed a highly durable hydrogel-based ECG electrode (Figure 6D; Table 1), which successfully solved this problem. It was then cut in a hydrogel/parylene bilayer into a serpentine design to ensure overall stretchability. Finally, the mixture was transferred onto the skin. The moulded e-tattoos have minimal/no gap at the interface with the skin, resulting in significantly lower interfacial impedance in the range of 20–1,000 Hz than gold standard medical wet Ag/AgCl electrodes do; moreover, owing to the presence of the modifier, the electrodes can be used for extended periods of time in excess of six months with consistently high signal fidelity. Transition metal disulfides (TMDs) in layered materials are intrinsically flexible, nearly invisible ($\sim <1$ nm) in thickness, yet highly adaptable, providing high compliance at the skin–device interface for e-tattoo electrodes. On this basis, Kirreev et al.¹⁷ developed an e-tattoo (Figure 6E; Table 1) that combines versatility and high stability and applied it to ECG monitoring. The electrode consists of platinum diselenide and platinum telluride, two types of platinum-based TMDs with large-scale, high-quality growth properties down to 400 °C, as well as high biocompatibility. The two materials were grown directly on Kapton or Silicon dioxide/Si wafers; subsequently, they were transferred to an ultrathin polymethylmethacrylate layer to form a temporary tattoo. The electrode outperforms conventional graphene e-tattoos and is comparable or even better than medical Ag/AgCl gel electrodes in several aspects, including conductivity, skin contact, and electrochemical impedance.

Textile-based electrodes

As intrinsically flexible materials, textile-based materials can be compounded with inherently rigid electronic devices to become textile-based electrodes, obtaining textile-based electrodes with intrinsic breathability, comfort, and wearability for long periods of time (Figure 7).¹⁴⁸ In these electrodes, the fiber acts as the basic constituent unit acts as a skeleton, but usually, it does not have conductivity, so how to make the fabric possess excellent electrical conductivity is the key. Current textile-based electrodes can be divided into two main categories: i) on the surface of the fabric through bonding, coating, and other methods to make it conductive and ii) using conductive fibers or yarns through textile technology to form fabrics.^{14,173,174} The first type usually has better conductivity, but the electrodes are more unstable in long-term use. Typically, the electrodes generate heat accumulation between the interior and the skin during long-term operation, which affects the work of the elec-

trodes after they accumulate heat to a certain amount. For this reason, Peng et al.²⁴ developed a fabric-based electrode with high stability and accuracy (Figure 7A); in particular, the electrode has a special passive radiation refrigeration function, which can reduce the concentration of heat inside the electrode under prolonged use and further improve the stability for long-term monitoring. Specifically, researchers first obtained SEBS microfibers by using an ionic liquid as a spinning agent and then uniformly dispersed functional aluminum oxide particles with selective spectral reflectance by sol-gel electrospinning, which have both high emissivity in the mid-infrared far-infrared bands and highly efficient backscattering of UV–visible–near-infrared radiation, which endows the electrode with passive radiative cooling capability, in order to reduce the internal heat buildup of the electrode under prolonged use. Finally, Eutectic gallium–indium, a liquid metal with extremely high electrical conductivity and stretchability, was printed on the nonwoven fabric obtained by electrospinning, which gives the electrode a high degree of accuracy for a wide range of electrophysiological signals. Yang et al.²¹ developed a Janus textile ECG electrode with a multilayer composite structure, focusing on solving the problem of sweat accumulation at the electrode–skin interface (Figure 7B; Table 1). The hydrophilic layer is formed by a hydrolyzed polyacrylonitrile fiber membrane deposited on an aluminum foil substrate supplemented with a medical adhesive, which can absorb sweat from the skin surface and evaporate it quickly by taking advantage of the strong capillary force and large wetting area formed by hydrolyzed more hydrophilic groups (e.g., hydroxyls, and so forth); the hydrophobic layer consists of electrospun layers of PU fibers, and alkyls inhibit the reverse osmosis of water to prevent reflux; the electrode layer is obtained by depositing AgNWs through vacuum filtration and is responsible for collecting bioelectric signals. Owing to the common structure of these layers, the electrodes can autonomously expel sweat and prevent reflux, so that there is no significant change in the baseline deviation and T/R ratio of the ECG signals before and after sweating and there is no signal weakening under long-term wear. The second type is limited by the production process of conductive fiber, and the electrode conductivity is much lower. Mixing multiple conductive materials is one of the solutions, but the problem of dispersion difficulty often occurs. Chih-Wei Chiu's team^{22,23} used a comb-shaped styrene-maleic anhydride -polyetheramine polymer/polyisobutylene-b-poly(oxyethylene)-b-polyisobutylene (PIB-POE-PIB) triblock copolymer to assist in the dispersion of a variety of conductive materials (Figure 7C; Table 1), including CB/rGO/CNT/AgNP/PEDOT:PSS. The maleic anhydride of the former interacts with CB to promote dispersion by adsorption, and the amino group (-NH₂) in the poly(ether amine) forms a hydrogen bond with the oxygen-containing functional groups (e.g., hydroxyl, -OH, and carboxyl, -COOH) in the rGO; the POE of the latter provides the oxygen atoms interacting with the AgNPs to form an ionic–dipolar interaction, and the PIB utilizes the hydrophobicity to assist in the dispersion of CB/rGO/CNTs. The electrodes obtained by the team all have excellent signaling capabilities, ensuring low impedance, high fidelity, and high stability of the electrodes. In addition, the direct fabrication of fibers from purely conductive materials is highly feasible. Taylor et al.⁸⁰

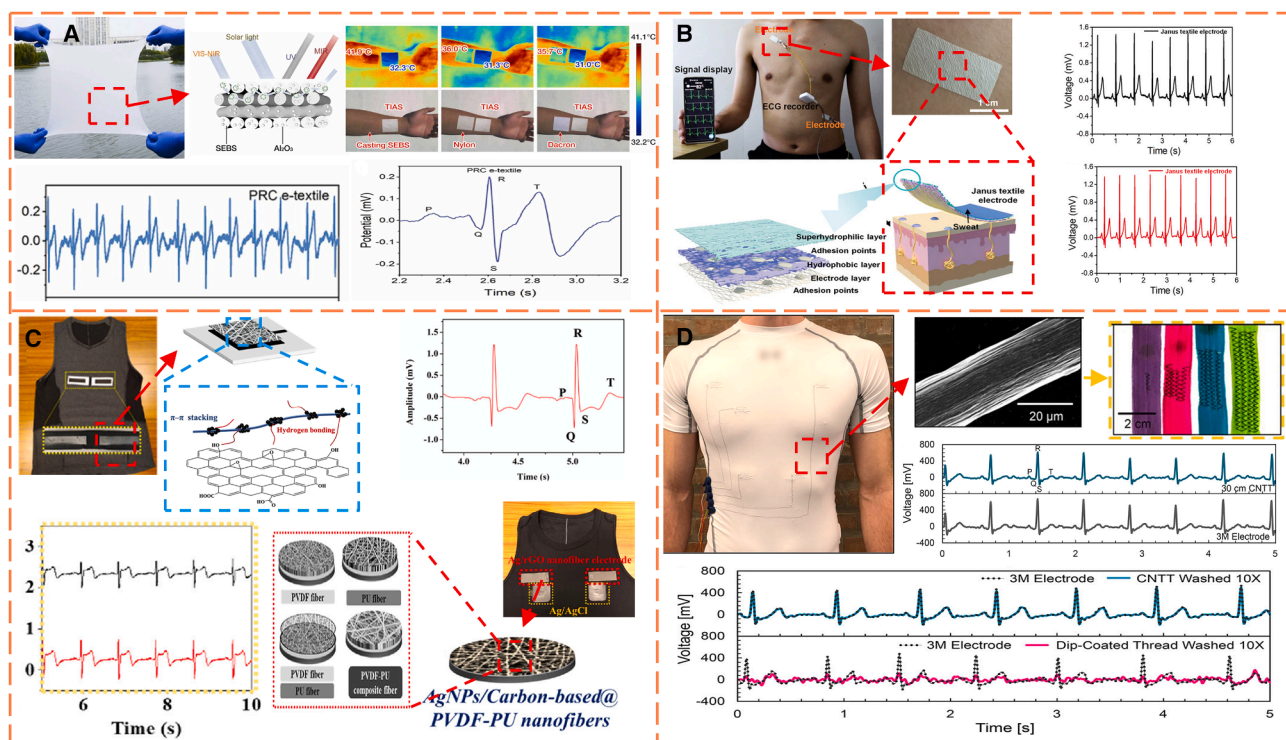


Figure 7. Advanced textile-based ECG electrodes

(A) Electrode with passive radiative cooling, the digital picture of the electrode is on the left side, the schematic of the electrode structure is on the upper side of the center, the schematic of the radiative cooling is on the upper side of the right side, and the acquired ECG signals are on the lower side. Reproduced with permission from²⁴; copyright 2023 Elsevier Ltd.

(B) Janus textile ECG electrode with a multilayer composite structure, with a digital photo of the electrode on the upper left side of the human body, a schematic of the electrode structure is on the lower left side, and the acquired ECG signal is on the right side. Reproduced with permission from²¹; Copyright 2022 Wiley-VCH.

(C) Electrodes of Chih-Wei Chiu's team. The upper left side is the digital photo of a comb-shaped SMA-M1000 electrode for assisted dispersion, the upper middle side is the schematic diagram of the electrode, and the upper right side is the ECG signals collected. Reproduced with permission from²²; Copyright 2021 American Chemical Society. PIB-POE-PIB triblock copolymer-assisted dispersion electrode is on the lower side, digital photo and schematic of the electrode is on the lower right side and the lower middle side, and the ECG signal acquired is on the lower left side. Reproduced with permission from²³; Copyright 2022 American Chemical Society.

(D) Pure CNT electrode with digital photo of the electrode is on the upper left side, the schematic diagram of the electrode is on the upper middle side, and the ECG signal acquired is on the upper right side and the lower side. Reproduced with permission from⁸⁰; Copyright 2021 American Chemical Society.

developed pure CNTs fibers and further fabricated them into electrodes with metal-level conductivity (Figure 7D; Table 1). Specifically, monofilament carbon nanotube fibers (CNTFs) were obtained by dissolving them in chlorosulfonic acid and spinning with acetone as the solidifying bath solution, which led to CNTs that maintained an average diameter of 1.8 nm, an average length-diameter ratio of 4,100, and a high degree of crystallinity with axial alignment properties, thus achieving an ultrahigh electrical conductivity of 6.6 ± 0.7 MS/m. Subsequently, CNT sewing threads were fabricated in groups of 21 and finally sewn into stretchable textiles, resulting in excellent ECG electrodes with low interfacial impedance, high chemical stability and durability, and ECG signal quality comparable to that of commercial ECG electrodes.

Microarray electrodes

Microarray electrodes (Figure 8) (MAEs), which evolved from cellular electrophysiology analysis techniques, contain unique symmetrically aligned array structures, recording electrophysio-

logical activities from cells or tissues and specific changes in the heart and brain in diseased states, and so forth, and have developed significantly in recent years.¹⁷⁵ MAEs are categorized into rigid MAEs (rMAEs), which are typically fabricated on rigid substrates such as glass, Si, or indium tin oxide coated glass; and flexible MAEs (fMAEs), which use polymer substrates such as parylene, PI, or PDMS. MAEs are valuable in biomedical research, as they can be used not only as sensors to detect chemicals and biomolecules, but also as electrodes for electrophysiological studies and drug screening. Park et al.²⁶ developed a flexible and stretchable graphene film electrode on biocompatible liquid crystal polymer (LCP) films via laser processing technology (Figure 8A; Table 1). The fabrication of the electrodes was accomplished via precise control of the laser scribing parameters, pattern transfer, and printing to take advantage of the intrinsic structural properties of the LCP source material, and the transfer printing technique allows the patterned laser-induced graphene (LIG) films to be applied to a variety of receiving substrates, efficiently transforming them into

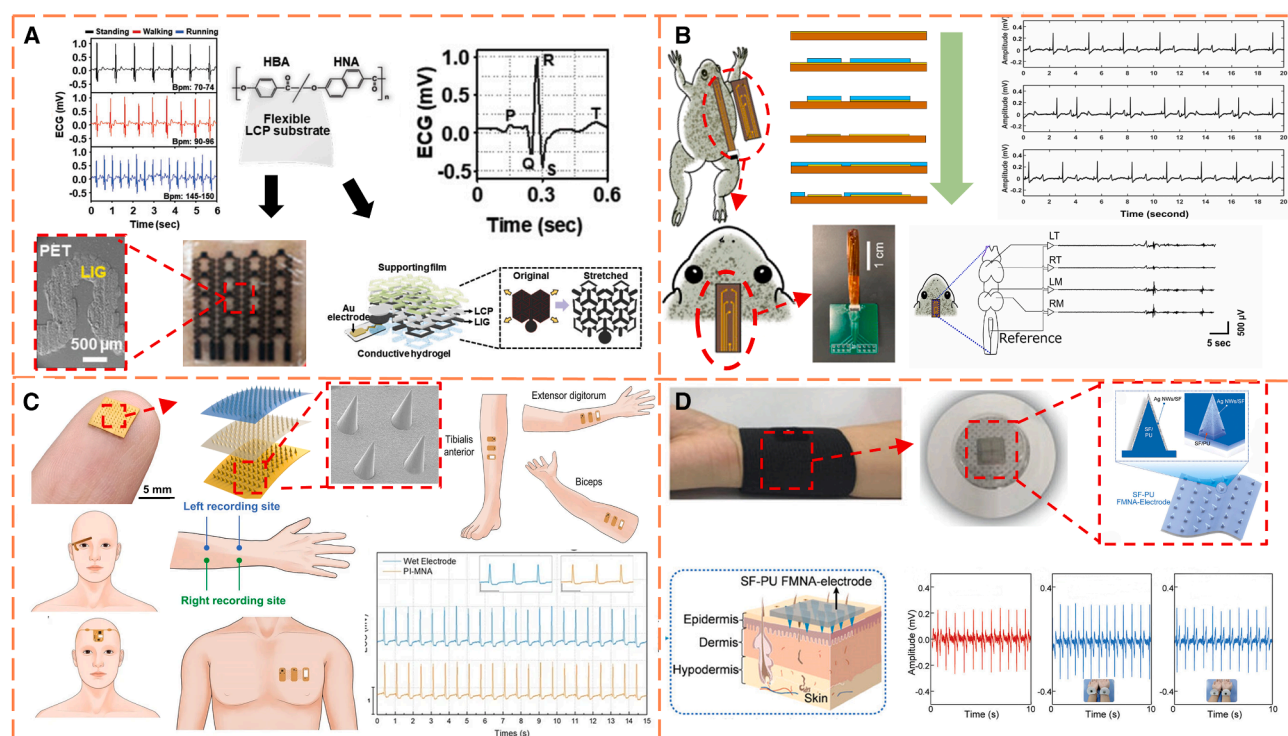


Figure 8. Advanced microarray ECG electrodes

(A) Array electrode with kirigami design based on laser processing technology, the basic material of the electrode is in the middle, the digital photo of the LIG array electrode and the microstructure is on the lower left side, the collected ECG signals is on the upper left side and the upper right side, and the schematic diagram of the electrode is on the lower right side. Reproduced with permission from²⁶; Copyright 2023 Authors.

(B) Microarray electrodes used for the real-time recording of electrophysiological signals in the African claw toad (*Xenopus laevis*), with digital photos of the electrodes on the lower left side, and the test schematic is on the lower right side, the manufacturing method of the electrode is on the upper left side, and the collected ECG signals are on the upper right side. Reproduced with permission from²⁵; Copyright 2022 Elsevier.

(C) Polyimide microneedle array (PI-MNA) surface biopotential dry electrode based on a PI substrate. The upper left side shows a digital photo of the electrode, the upper middle side shows a schematic of the electrode structure and an SEM photo of the electrode, the types of signals that can be detected by the electrodes and where the electrodes are placed are on the upper right side, lower left side and lower center, and the right side shows the collected ECG signals. Reproduced with permission from²⁶; Copyright 2022, Author(s).

(D) Silk protein (SF)-PU-mediated hybrid microneedle array electrode. The digital photo of the electrode is on the upper left side and upper middle side, the upper right side, and the lower left side is a schematic diagram of the electrode structure, and the lower right side is the collected ECG signal. Reproduced with permission from²⁷; Copyright 2024 Wiley-VCH.

LIG-based printed circuit boards and encapsulating them with flexible elastomers for added protection and mechanical strength. The resulting electrodes have a variety of properties, including high electrical conductivity, biocompatibility, appropriate mechanical strength, and versatility with specific biomolecules, making LIG-based electrodes highly advantageous in the field of signal detection. In particular, researchers have also demonstrated an LIG electrode for skin attachment that employs a kirigami design to adapt to the deformation of the skin in different behavioral states and to maintain stability and accuracy in electrophysiological signal monitoring. The electrode can transmit high-resolution and accurate ECG signals to mobile devices in real time, providing an ECG monitoring solution with a large contact area with the skin and high signal quality. Xia et al.²⁵ developed an integrated bioelectronic sensing system capable of simultaneously monitoring ECGs and electroencephalograms (EEGs) (Figure 8B) for real-time recording of electrophysiological signals in *Xenopus*. The electrodes used MAEs

with gold electrodes arranged on a flexible polyimide membrane. The electrodes were obtained by depositing an Au/chromium layer on a PI substrate, using photolithography and wet etching processes to form electrode patterns, and finally encapsulating the electrode tracks with a hardened photoresist layer. Based on the brain structure of *Xenopus*, researchers have designed 4-channel EEG recording electrodes placed above the cerebral cortex with reference electrodes above the cerebellum. The features of the electrodes include their excellent biocompatibility and flexibility to achieve uniform and safe contact with *Xenopus* skin for noninvasive multimodal electrophysiological monitoring with good spatial resolution. The size of the MAE membrane was designed to be very small to minimize disturbance to the animal while maintaining high-quality ECG and EEG recordings. Additionally, based on a PI substrate, Li et al.²⁸ developed a PI-microneedle array (MNA) surface biopotential dry electrode for electrophysiological signal monitoring (Figure 8C). By depositing metal layers (Ti and Au) and a conductive polymer layer

(PEDOT:PSS) on the MNAs obtained from the substrate, these electrodes gained sufficient adhesion and signaling ability; when attached to the skin, the PI-MNA electrodes could penetrate the stratum corneum to reduce contact impedance while avoiding contact with deeper tissues, such as nerves and blood vessels, and thus achieve high-quality monitoring. With record-breaking low contact impedance ($\sim 1/250$ times that of standard electrodes), the electrode can easily obtain accurate data over an extremely long period of time (~ 44 nights, >8 h per night), making it an ideal replacement for standard clinical electrodes. However, Wang et al.²⁷ reported that the low conductivity or poor biocompatibility present in the commonly used MNAs of PI, PDMS, and Ecoflex hindered the development of these types of MNAs; thus, the team used silk protein (SF) as a material to improve this situation by mediated mixing with SF-PU (Figure 8D; Table 1). Specifically, a negative mold was first obtained by casting PDMS on a commercial silicon substrate, and then SF was obtained by simple immersion in an alkaline solution with Lithium bromide (LiBr) solution. AgNWs obtained by chemical reduction and a PU solution were poured into the mold for mesoscopic reconstruction, and ethanol vapors were used to promote the β -crystalline transformation of SF to increase the overall stability of the SF crystalline structure. Finally, the electrode surface was modified with PEDOT:PSS to further improve the conductivity. The resulting electrode overcomes the shortcomings of commonly used substrates such as excellent biocompatibility, a high signal-to-noise ratio, low contact impedance, and many other advantages, and has excellent application prospects.

CHALLENGE AND OUTLOOK

The novel wave of innovations brought by the development of flex electronics has had a wide range of applications in health care¹⁷⁶; in particular, the breakthrough of FDEs for ECG monitoring has already led to desirable application prospects. However, their inherent drawbacks limit areas such as the monitoring environment, stability, signal quality, signal transmission, and power supply; thus, challenges and opportunities remain. Here, we depict the shortcomings of the electrodes and provide possible directions to address them.

Signal quality and stability under special conditions

Electrodes typically need to maintain good contact with the skin during operation to ensure high-quality, stable signal transmission; in particular, ECG electrodes need to be tightly adhered to the skin to minimize contact impedance and noise. Generally, they work for a short period of time and under relatively gentle monitoring conditions. However, the low adhesion of the electrodes and the mismatch of the Young's modulus with the skin may lead to slippage or even detachment of the electrodes, which could cause great fluctuations in the signal quality of the electrodes, including during long-term monitoring, exercise, and wet environments. Besides, since the performance of the electrodes will deteriorate after long-term use, it is also very necessary to analyze the degradation mechanism of the electrodes. Therefore, regarding the electrode degradation mechanism of e-skin/tattoo electrodes during long-term use, it can be

discussed from the following three aspects. a) Micro-cracks within the electrode conductive network, which result from long-term mechanical deformation, lead to irreversible electrode conductive-path damage. b) Long-term sweat accumulation between the skin-electrode interface and/or skin micromotion could decrease the adhesion to skin. c) The electron devices tend to be degraded by long-term exposure to e sweat, especially for on-chip amplification circuits or metal traces without encapsulation, where ionic migration or delamination can occur. When it comes to the textile electrodes, there are two electrode degradation mechanisms. a) Similar to an e-skin/tattoo electrode, micro-cracks caused by mechanical deformation could damage the conductive network, especially at stress-concentration zones in the knit structure. b) After long-term wearing and repeated laundering, the adhesion between the conductive coating and the textile changes, leading to irreversible damage of interface conductivity, even causing skin irritation. In terms of microarray electrodes, after long-term use, the electrode degradation mechanisms comprise the following three types. a) Long-term use introduces micro angular shifts in the micro-needles/micropillars, and this kind of array-level fatigue may cause subtle displacement or blunting. b) Localized corrosion triggered by long-term working in body fluids drives a permanent rise in electrode impedance. c) For several hollow arrays, tissue fluid or cell debris clogs the channels and blocks signal transmission. Furthermore, the reproducibility and stability of the electrodes during long-term use may be affected by the oxidation of the electrode material and swelling of the hydrogel in water. These electrode degradation mechanisms severely limit the long-term use under current design paradigms, which need to be taken into consideration during the electrode design process. Furthermore, the reproducibility and stability of the electrodes during long-term use may be affected by the oxidation of the electrode material and swelling of the hydrogel in water. The development of submicron self-adhesive films with a skin-like Young's modulus can alleviate the situation to a certain degree, but it is still unable to solve the problem completely.¹⁷⁷

Therefore, future electrodes should minimize the effects of oxidation and swelling while having a Young's modulus like that of skin and high adhesion. In order to lower the effects, future electrodes need to utilize antioxidant and anti-swelling materials. For skin-like Young's modulus, the electrodes should be made of highly flexible materials and possess micro- or nano-conformable structure. Lastly, for high adhesion, there are three common methods to achieve this goal. a) Drawing on biomimicry, micro- and nano-structures, such as octopus-like suction cups, remora vacuum cavities, or gecko setae, could be designed by structural engineering. By leveraging physical principles such as pressure differentials and capillary forces, stable interfacial adhesion can be achieved. b) Through chemical methods, specific materials can be treated at their surfaces or structures to expose functional groups. This enables the formation of strong interfacial adhesion via intermolecular interactions such as hydrogen bonds, van der Waals forces, and electrostatic adsorption between molecules at the skin-electrode interface. c) Employing a combination of physicochemical methods alongside biomimetic structural design and functional group design

in materials to achieve optimal adhesion performance. Future electrodes should adopt one of them to maintain stable high adhesion. Importantly, the skin-matched low modulus ensures that the adhesive micro-structures remain fully engaged with the epidermal ridges during micro-slippage, preventing local stress concentration and delaying interfacial crack propagation. Moreover, electrodes that possess these characteristics are highly expected to take into account the signal quality and stability of long-term monitoring, in-motion monitoring, or wet environment monitoring.

Wireless data transmission and limitations of the power supply

In the field of ECG monitoring, wireless data transmission eliminates the limitations of traditional data lines and provides convenience for portable and nonclinical monitoring. For both single-lead ECG and multi-lead ECG, signals are collected centrally by electrodes and transmitted in packaged form via Bluetooth, microwave, or other ways to the analysis terminal device (mobile phone, electrocardiograph, and so forth). During this process, the site contributes one potential; the on-chip differential amplifier then forms the ECG vector against the companion reference (either a dedicated reference electrode in single-lead mode or the WCT in 12-lead mode) before wireless transmission. According to the lead configuration, the transmitted signals contain either information carrying a reference electrode (single-lead) or multiple signal sets without a reference electrode (12-lead), ensuring accurate and effective signal transmission. Although a minimum of two nodes (signal and reference) are required to form one ECG channel, integrating an ultra-low-power Application-Specific IC directly at the electrode site eliminates the long shielded cable, reduces motion artifacts, and allows modular lead expansion (e.g., 12 independent patches communicating to one hub). Thus, 'per-electrode wireless' refers to the replacement of wired harnesses by localized, short-range wireless links rather than to a true single-node measurement. However, it typically uses rigid commercial integrated circuit (IC) chips, which increases the foreign body sensation of the electrodes and reduces their comfort and portability. In addition, the high energy consumption of the communication module (for example, conventional Bluetooth technology consumes 1–5 mW of power) exceeds the power supply capability of most thin-film batteries (<1 mW), and the rigidity of conventional thin-film batteries is also present, which further restricts the portability of the electrodes. Recent research has shown that signal collection and transmission can be integrated into fibers, avoiding the disadvantage of higher rigidity of the chips of traditional von Neumann architecture electrodes¹⁷⁸; the development of self-powered technologies has gradually filled the latter gap, with nanogenerators that can carry out additional energy storage and thus provide the required electrical energy. More importantly, by using flexible electronics technology, nanogenerators can be self-powered in combination with electrodes without the need for an external power source, which greatly improves the portability of the overall system.^{179–181} Therefore, future electrodes can realize wireless data transmission in the presence of flexible self-powered devices, which has excellent application value.

DISCUSSION

In this review, we investigated the latest research advances in FDEs for ECG monitoring, which have revolutionary potential in medical diagnosis and daily health care. With the rapid development of wearable technology, FDEs have found a new favor in the field of ECG monitoring because of their excellent comfort and stability for long-term wear. This article reviews the working principles of FDEs for ECG monitoring and their design strategies, including conductive materials, flexible matrix materials, and electrode structures, and analyses advanced technologies such as e-skin/e-tattoo electrodes, textile-based electrodes, and microarray electrodes. These innovative electrode designs not only improve the quality of ECG signal acquisition but also offer the possibility of monitoring in dynamic and multisenario situations. However, despite these encouraging advances, the development of FDEs faces challenges. Maintaining signal quality and stability during prolonged monitoring, during motion, and in wet environments are key challenges in current research. In addition, limitations in wireless data transmission and power supply need to be addressed. Future research needs to find solutions in the cross-fertilization of multiple disciplines, including materials science, electrical engineering, and biomedical engineering, to achieve more comfortable, stable, and intelligent ECG monitoring systems. In the future, we anticipate the development of FDEs toward integration, intelligence, and self-powering. The development of novel electrodes with highly efficient conductive networks, textile structures, or microneedle array structures, and high skin compliance by combining a variety of materials and structural designs will be the key to achieving this goal. In addition, the utility and user-friendliness of the electrodes can be further enhanced by mimicking the physiological properties of human skin, such as self-healing ability and adaptability. As these technologies mature, FDEs are expected to play an even more important role in the prevention, diagnosis, and treatment of heart disease, highlighting the importance of global heart health monitoring.

Limitations of the study

In this review, flexible dry electrodes for ECG monitoring were analyzed according to their working principles, design strategies, and advanced applications. However, several typical types of electrodes were analyzed. Other promising flexible electrode types, such as those based on foam, article, are not discussed in depth, which may limit the breadth of the coverage. Besides, this review may not encompass all recent studies, especially those published in non-English journals or in preprint repositories. By addressing these limitations in future work, there would be great progress in the development of flexible dry electrodes.

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AUTHOR CONTRIBUTIONS

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DECLARATION OF INTERESTS

The authors declare no conflict of interest.

DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES IN THE WRITING PROCESS

During the preparation of this work, the author(s) used DEEPSEEK in order to polish the article. After using this tool or service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

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